

Object Oriented Programming
Midterm Exam SY 2nd sem

Select the best answer:

1. What is the primary reason for using files instead of RAM for long-term data storage?
 - A. Files are faster than RAM
 - B. Files are non-volatile and retain data even after the computer is turned off
 - C. Files are cheaper to use than RAM
 - D. Files do not require any storage space
2. Which of the following is an example of a binary file?
 - A. .txt
 - B. .py
 - C. .jpg
 - D. .log
3. What file permission mode in Python allows you to add data to the end of an existing file without overwriting its contents?
 - A. 'r'
 - B. 'w'
 - C. 'a'
 - D. 'x'
4. What Python command is used to open a file named `data.txt` for reading?
 - A. `open("data.txt", "w")`
 - B. `open("data.txt", "a")`
 - C. `open("data.txt", "r")`
 - D. `open("data.txt")`
5. Which function reads the entire content of the file as a single string?
 - A. `readlines()`
 - B. `read()`
 - C. `readline()`
 - D. `readall()`
6. What is the purpose of the `close()` function after file operations in Python?
 - A. To hide the file from the user
 - B. To free up system resources tied to the file
 - C. To delete the file
 - D. To lock the file
7. Which method reads one line from a file at a time?
 - A. `read()`
 - B. `readlines()`
 - C. `readline()`
 - D. `readchar()`
8. What happens when you open a file with mode 'w'?
 - A. The file is opened for reading only
 - B. The file is opened and existing data is preserved
 - C. The file is opened and existing data is overwritten
 - D. The file cannot be opened if it doesn't exist
9. Which of the following is the correct way to loop through all lines in a file named `example.txt`?
 - A. `for line in open("example.txt", "w"):`
 - B. `for line in open("example.txt", "r"):`
 - C. `for line in open("example.txt", "a"):`
 - D. `for line in open("example.txt"):`
10. Which function reads all lines in a file and returns them as a list?
 - A. `readlines()`
 - B. `read()`
 - C. `readline()`
 - D. `readlist()`
11. What is the main purpose of inheritance in programming?
 - A. To make the program slower
 - B. To reuse existing code
 - C. To reduce the size of the file
 - D. To increase file storage
12. In Python, what keyword is used to refer to the parent class inside a child class?
 - A. `parent()`
 - B. `super()`
 - C. `base()`
 - D. `inherit()`
13. Which of the following best describes Single Inheritance?
 - A. A child class inherits from multiple parent classes
 - B. A child class inherits from a single parent class

- C. A parent class inherits from a child class
 - D. Two unrelated classes share the same methods
14. Which of the following types of inheritance involves a chain of classes, like Class A → Class B → Class C?
- A. Multiple Inheritance
 - B. Hierarchical Inheritance
 - C. Multilevel Inheritance
 - D. Hybrid Inheritance
15. What is the advantage of using method overriding?
- A. To delete the parent class
 - B. To allow the child class to extend or modify inherited methods
 - C. To lock the parent class methods
 - D. To prevent changes in the child class
16. Which statement about polymorphism is TRUE?
- A. It allows one function name to perform different tasks depending on context
 - B. It prevents method overriding
 - C. It limits code flexibility
 - D. It forces all classes to use the same methods exactly
17. Which of the following is an example of polymorphism in Python?
- A. Using the `len()` function on both a list and a string
 - B. Using only one type of data structure
 - C. Preventing a child class from changing a parent method
 - D. Declaring variables without initializing
18. What type of inheritance involves one child class inheriting from multiple parent classes?
- A. Multiple Inheritance
 - B. Single Inheritance
 - C. Hybrid Inheritance
 - D. Multilevel Inheritance
19. Which inheritance structure has one parent class with multiple child classes?
- A. Single Inheritance
 - B. Hierarchical Inheritance
 - C. Hybrid Inheritance
 - D. Multilevel Inheritance
20. What benefit does the `super()` function provide in Python inheritance?
- A. Requires writing the parent class name repeatedly
 - B. Prevents reuse of code
 - C. Simplifies calling parent class methods
 - D. Disables inheritance features
21. What is a Class in Python?
- A. A collection of random values
 - B. A user-defined data structure that binds data and methods into a single unit
 - C. A function that performs calculations
 - D. A variable used to store user input
22. What is an Object in Python?
- A. A built-in data type
 - B. A block of code that performs actions
 - C. An instance of a class with its own identity, state, and behavior
 - D. A keyword for creating classes
23. How do you access an instance attribute of an object in Python?
- A. `object.attribute`
 - B. `class.attribute`
 - C. `self.attribute()`
 - D. `get.attribute()`
24. Which keyword is used to define a class in Python?
- A. `define`
 - B. `object`
 - C. `class`
 - D. `structure`

25. What is the role of a constructor in Python?
- A. To delete an object
 - B. To create a class
 - C. To initialize an object's attributes
 - D. To define a function
26. What is a default constructor in Python?
- A. A constructor with parameters
 - B. A constructor automatically provided if no constructor is defined
 - C. A constructor that deletes data
 - D. A constructor that takes user input
27. What distinguishes a parameterized constructor from a non-parameterized constructor?
- A. It does not initialize any values
 - B. It uses random values by default
 - C. It accepts arguments when creating an object
 - D. It only works with lists
28. What is the purpose of the self keyword in Python?
- A. It refers to the parent class
 - B. It allows accessing the current object's attributes and methods
 - C. It declares a global variable
 - D. It creates a static method
29. Which of the following is TRUE about instance variables?
- A. They are shared across all instances
 - B. Each object has its own copy of instance variables
 - C. They are accessible only through class names
 - D. They are deleted when the class is declared
30. In the provided Circle class example, how is the radius set to a new value?
- A. `circle.radius = new_value`
 - B. `circle.new_value()`
 - C. `circle.setRadius = new_value`
 - D. `circle.changeRadius()`
31. What does it mean when we say **strings are immutable** in Python?
- A. Strings can change their characters anytime
 - B. String length can be doubled automatically
 - C. Strings cannot be changed after they are created
 - D. Strings can be modified using their index positions
32. How do you create a string in Python?
- A. By using curly braces {}
 - B. By using single, double, or triple quotes
 - C. By using square brackets []
 - D. By using parentheses ()
33. What is the result of "Program"[3:6]?
- A. 'Pro'
 - B. 'gra'
 - C. 'ram'
 - D. 'ogr'
34. Which method is used to convert all characters in a string to **lowercase**?
- A. `lower()`
 - B. `upper()`
 - C. `title()`
 - D. `capitalize()`
35. Which string method **capitalizes** only the first letter of a string?
- A. `title()`
 - B. `capitalize()`
 - C. `upper()`
 - D. `swapcase()`

36. What is the result of "A" * 4 in Python?
A. 'AAAA' B. 'A4' C. '4A' D. 'A A A A'
37. Which method removes both **leading and trailing spaces** from a string?
A. lstrip() B.rstrip() C. strip() D. trim()
38. What does the find() method return?
A. The number of times a character appears B. The length of the string
C. The index of the first occurrence of a character D. True or False
39. Which method converts all characters in a string to **uppercase**?
A. upper() B. lower() C. swapcase() D. title()
40. What does the split() method do?
A. It reverses the string
B. It joins two strings
C. It splits the string into a list of words or substrings
D. It removes all spaces in the string
41. What is an **exception** in Python?
A. A way to store data
B. A signal that an error has occurred during program execution
C. A block of comments in the code
D. A method to create functions
42. Which Python keyword is used to handle exceptions?
A. try B. check C. error D. validate
43. What is the purpose of the except block in Python?
A. To execute code only when there is no error
B. To specify the error-handling code when an exception occurs
C. To skip the execution of the program
D. To declare a function
44. Which of the following correctly handles **ZeroDivisionError** in Python?
A. try: 5/0 except ZeroDivisionError: print("Error")
B. check: 5/0 except ZeroDivisionError: print("Error")
C. try 5/0: except ZeroDivisionError print("Error")
D. validate 5/0 error ZeroDivisionError print("Error")
45. Which of the following is **not** a built-in Python exception?
A. IndexError B. KeyError C. NameError D. NumberError
46. What does the **finally** block do in Python exception handling?
A. It skips the execution if there is no error
B. It always executes, whether an exception occurs or not
C. It ends the program
D. It handles only keyboard interruptions
47. What will happen if you **do not handle** an exception in Python?
A. The program will continue without stopping
B. The program will display an error message and stop
C. The program will fix the error automatically
D. The program will ask the user to fix it

48. What is the output of the following?

```
try:
    print(10 / 0)
except ZeroDivisionError:
    print("Division by zero error")
```

- A. 0 B. Division by zero error C. Error D. Program crashes

49. Which block is **optional** in Python exception handling?

- A. try B. except C. finally D. None of the above

50. What is the correct way to handle **multiple exceptions** in Python?

- A. Use multiple `try` blocks
B. Use multiple `except` blocks under one `try` block
C. Use `check` statements
D. Use `validate` statements